3-D transient pressure / velocity compatibility

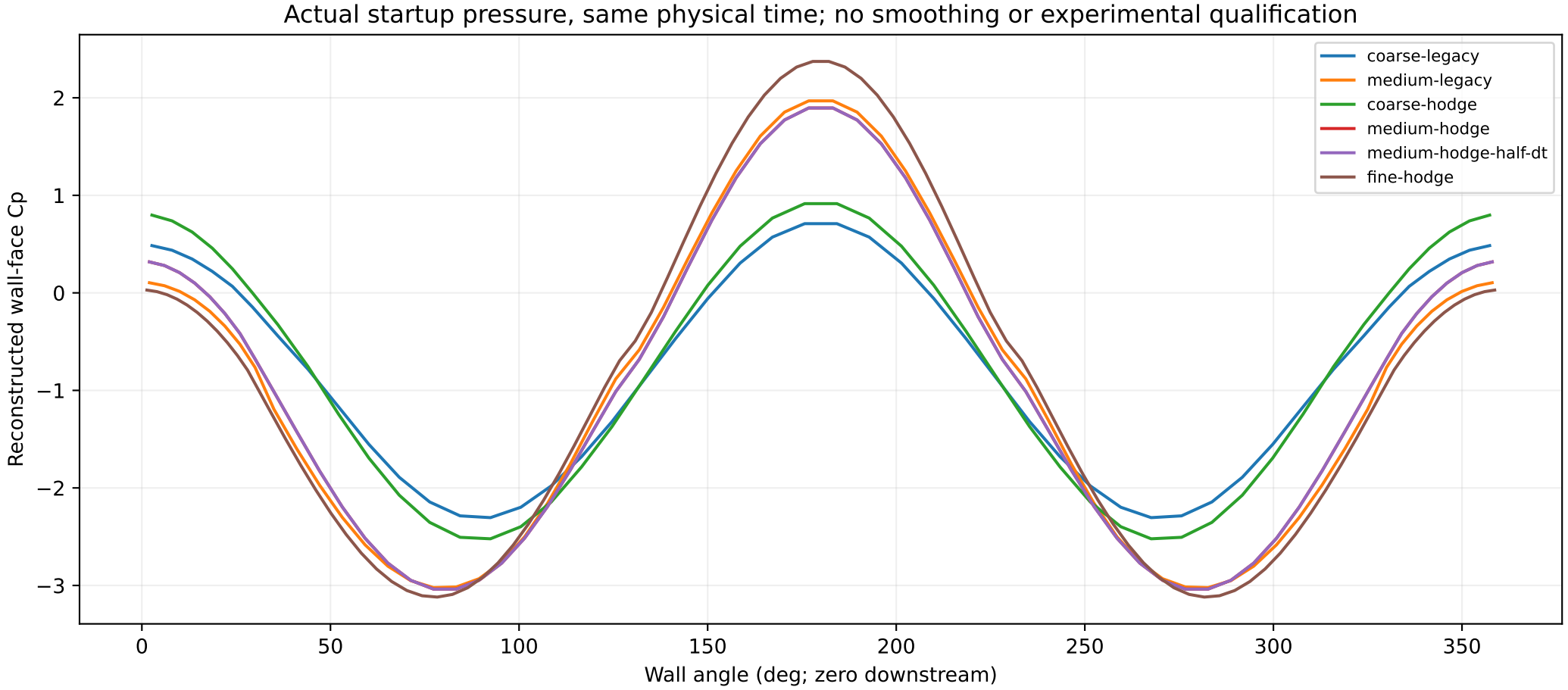
Development verification: physical Re3900 qualification remains FALSE.

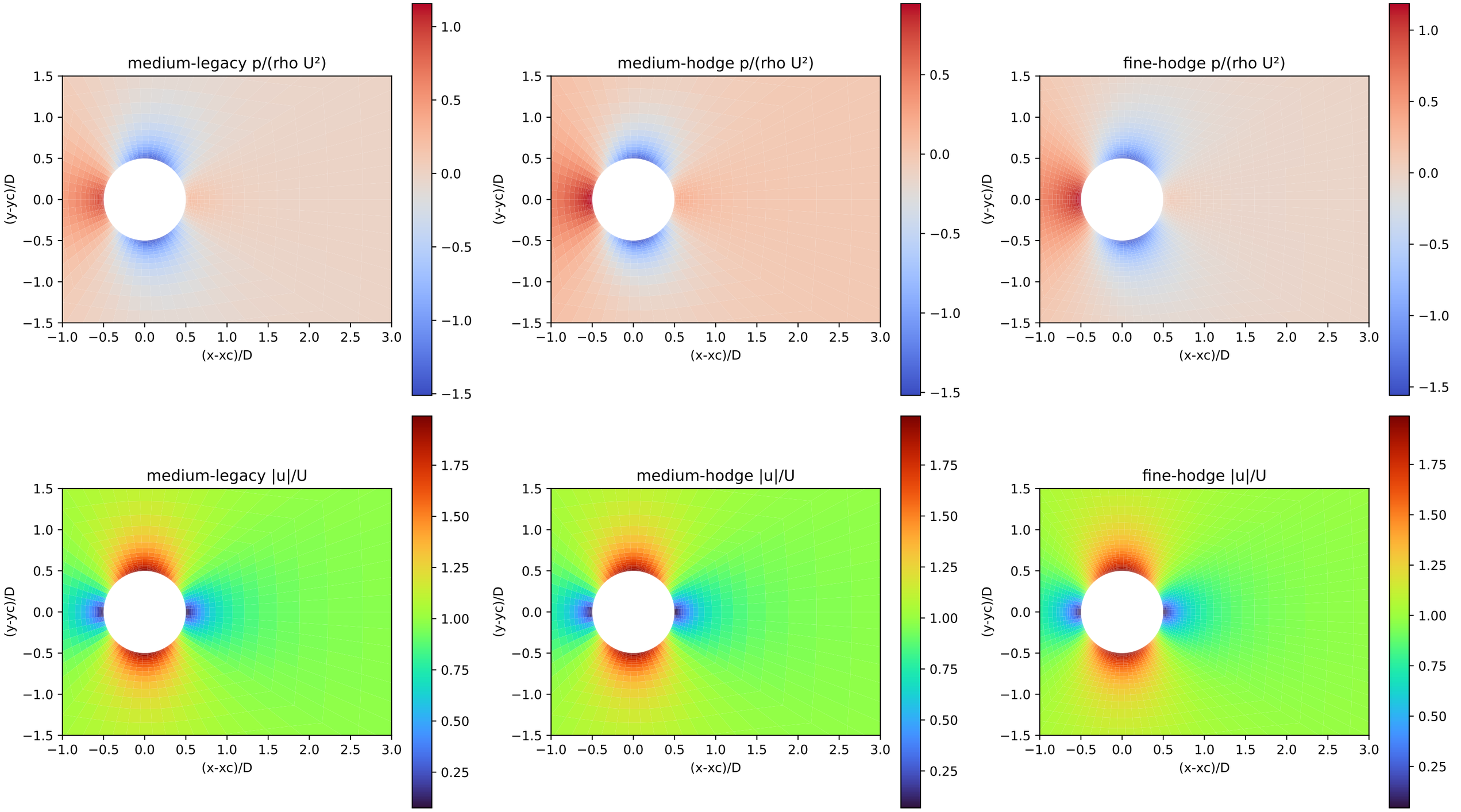
The original uniform-cell startup sets wall mass flux to zero, while
using homogeneous pressure Neumann data for its impulse.
The new opt-in startup solves a dt-independent velocity-potential
projection with the actual wall-normal impulse as Neumann data.
Shared-face pressure traction updates cell momentum conservatively.
Phi is not physical pressure; no elapsed time or force sample is added.

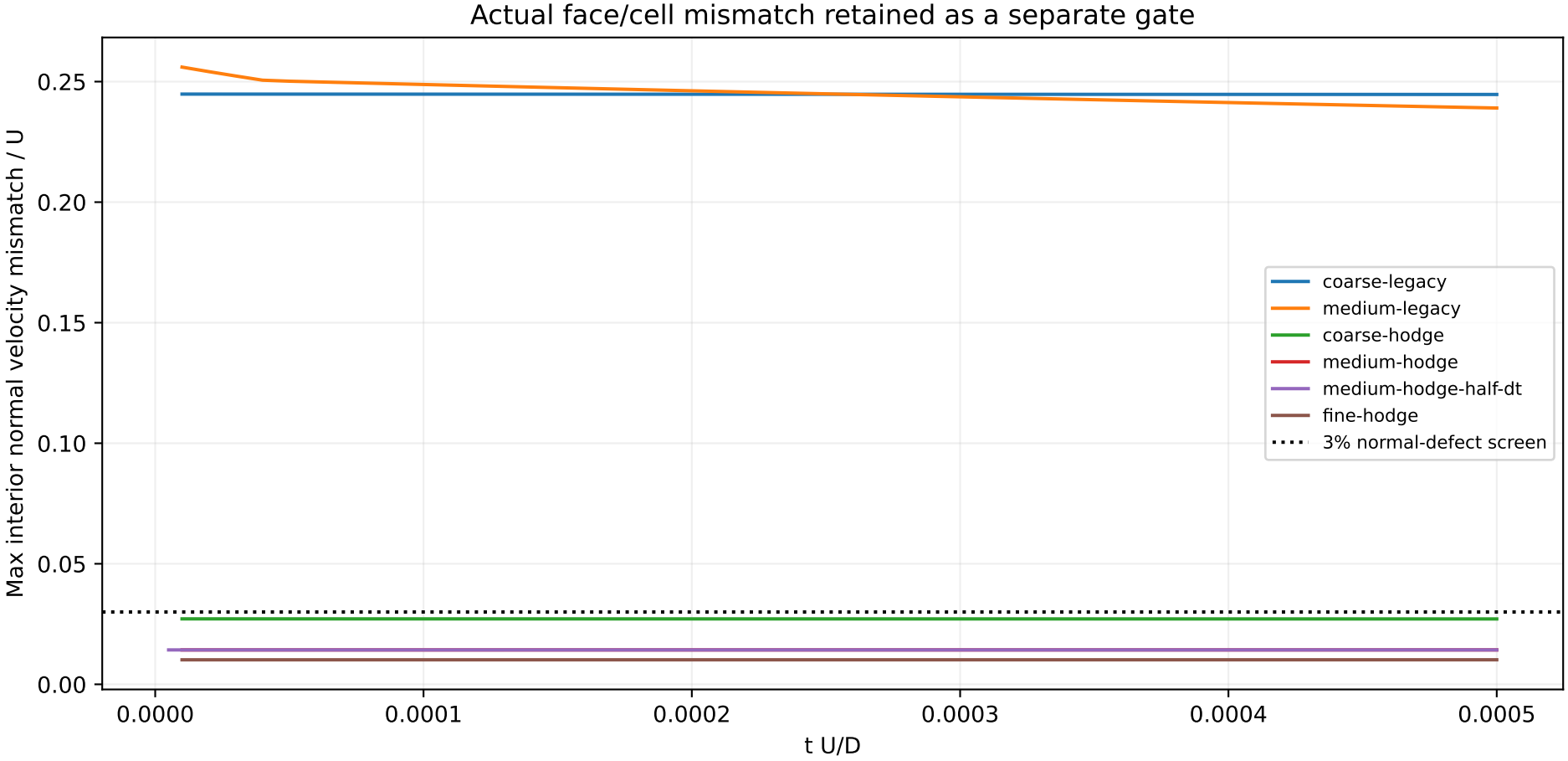
Affine manufacture: passed=True.
Medium normal defect legacy / Hodge: 0.239047 / 0.0142811 U.
Fine Hodge normal defect: 0.0101762 U.
Medium interpolated divergence: 3.87587 U/D.

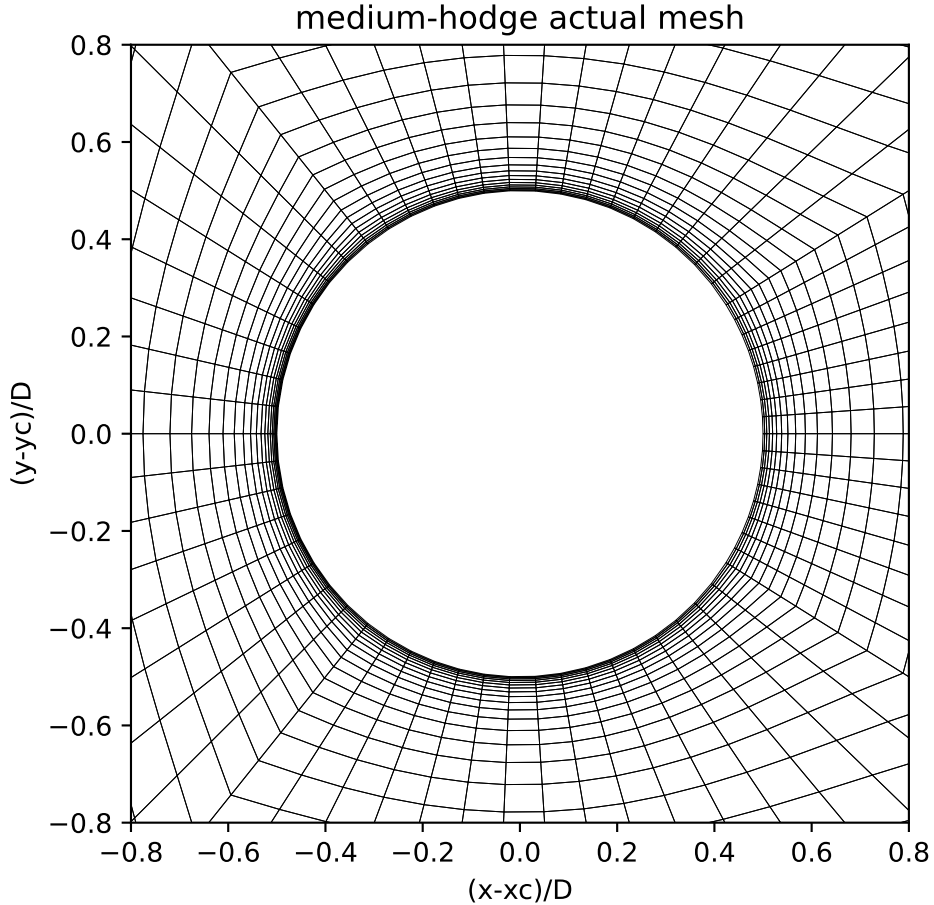
The normal-defect improvement does not make the cell and face
pressure operators identical. Strict cell divergence and startup
Cp/grid response are reported separately, including failures.
Old and Hodge starts have different initial cell fields; this is an
initial-condition correction, not an identical-start time comparison.
No velocity filtering, nonconservative reconstruction, or Cp fitting.

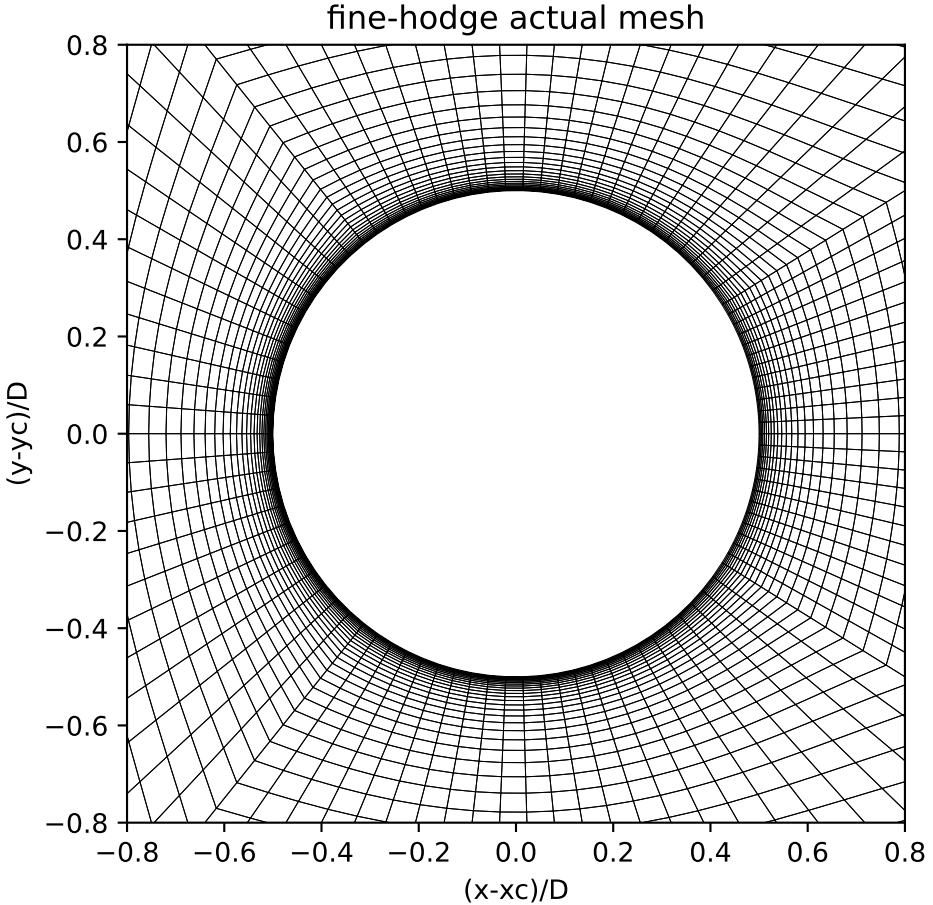
Ordinary time steps still use homogeneous pressure Neumann data.
Full momentum-compatible wall pressure and long-time Navier-Stokes
validation remain required. CPU only; no LES, CUDA, or SUBOFF
hydrodynamic qualification is claimed. See report.md / summary.json.


Actual startup pressure, same physical time; no smoothing or experimental qualification
Reconstructed wall-face Cp
Wall angle (deg; zero downstream)
coarse-legacy
medium-legacy
coarse-hodge
medium-hodge
medium-hodge-half-dt
fine-hodge
2
1
0
−1
−2
−3
0
50
100
150
200
250
300
350


medium-legacy p/(rho U²)
medium-hodge p/(rho U²)
fine-hodge p/(rho U²)
medium-legacy |u|/U
medium-hodge |u|/U
fine-hodge |u|/U
(x-xc)/D
(y-yc)/D


Actual face/cell mismatch retained as a separate gate
Max interior normal velocity mismatch / U
0.25
0.20
0.15
0.10
0.05
0.00
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
t U/D
coarse-legacy
medium-legacy
coarse-hodge
medium-hodge
medium-hodge-half-dt
fine-hodge
3% normal-defect screen


medium-hodge actual mesh
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
(y-yc)/D
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8
(x-xc)/D


fine-hodge actual mesh
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
(y-yc)/D
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8
(x-xc)/D